

**B-2-A**

Roll No.....

Total No. of Questions : 21]

[Total No. of Printed Pages : 4

**XIISZRJDF20  
1102-A  
CHEMISTRY**

Time : 3 Hours]

[Maximum Marks : 70

(Long Answer Type Questions)

5 each

1. Explain the terms :

- (i) Molarity
- (ii) Molality
- (iii) Mole fraction
- (iv) Normality
- (v) Parts per million

*Or*

What are colligative properties ? Derive the relation between the elevation in boiling point and molecular mass of a non-volatile solute in solution.

2. What is Corrosion ? Explain electrochemical theory of rusting.

*Or*

What is Electrolysis ? State and explain Faraday's laws of electrolysis.

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Turn Over

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3. Name the elements of nitrogen family. Based on the electronic configuration explain the valence of N and P. Give common oxidation states of nitrogen.

*Or*

How is sulphuric acid manufactured by contact process ? Why is sulphuric acid syrupy (viscous) ?

4. Write notes on the following :

- (i) Aldol condensation
- (ii) Cannizzaro's reaction
- (iii) Rosenmund reduction

*Or*

What are Carboxylic Acids ? Give any *four* general methods of preparation of carboxylic acid.

**(Short Answer Type Questions)**

**3 each**

- 5. What is difference between 'Schottky defect' and 'Frankel defect' ?
- 6. Show that for the reaction of first order half-life period is independent of initial concentration.
- 7. Explain the following terms in relation to colloidal solution :
  - (i) Brownian movement
  - (ii) Tyndall effect

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8. What is meant by unidentate and bidentate ligands ?
9. What are Alcohols ? How are they classified ?
10. Give any *three* oxidising properties of  $K_2Cr_2O_7$ .
11. What are essential and non-essential amino acids ? Give *one* example of each.
12. How is  $PCl_3$  prepared from white phosphorus ? How does it react with (i)  $SO_3$ , (ii)  $S_2Cl_2$  ?

(Very Short Answer Type Questions)

2 each

13. Define rate of reaction and rate constant.
14. What is Roasting ?
15. Why are transition metals called *d*-block elements ?
16. What is Diazotisation ?
17. How will you convert ethyl bromide into :  
(i) Ethyl alcohol  
(ii) Ethyl cyanide ?
18. Discuss Wurtz reaction.
19. What are Antibiotics ? Give example.
20. Distinguish between homopolymer and copolymer.

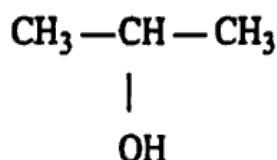
(Objective Type Questions)

1 each

21. (i) The fourteen types of space lattices are collectively called .....

( 4 )

- (ii) Colloidal dispersion of a liquid in a gas is called .....
- (iii) Carbylamine reaction is given by primary as well as secondary amines. (True/False)
- (iv) In aqueous solution trimethylamine is more basic than methylamine. (True/False)
- (v) What is Refining ?
- (vi) Give IUPAC name of :



- (vii) What is Vinegar ?
- (viii) Which of the following is not a thermoplastic ?
- |               |              |
|---------------|--------------|
| (A) Polythene | (B) Teflon   |
| (C) Nylon-6,6 | (D) Bakelite |
- (ix) Which of the following Vitamins is water soluble ?
- |               |               |
|---------------|---------------|
| (A) Vitamin E | (B) Vitamin K |
| (C) Vitamin A | (D) Vitamin B |
- (x) Equanil is .....
- |                          |                        |
|--------------------------|------------------------|
| (A) Artificial sweetener | (B) Tranquillizer      |
| (C) Antihistamine        | (D) Antifertility drug |

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